Table 10.5
Standard Test Methods for Asphalt Content of Hot Mix Asphalt

Standard	Description of Method	Notes
ASTM D 2172 AASHTO T 164	Asphalt Content by Solvent Extraction	Extracted aggregate can be used for gradation testing. Users must comply with hazardous material handling requirements.
ASTM D 4125 AASHTO T 287	Asphalt Content by Nuclear Method	Rapid turnaround on test results is possible. Aggregate is not available for gradation because asphalt is not removed. Special licensing requirements and safety procedures are required.
ASTM D 6307 AASHTO T 308	Asphalt Content by Ignition Method	Remaining aggregate can be used for gradation testing. Exercise caution with certain aggregates.

10.3.4 Maximum Theoretical Specific Gravity

Testing for maximum theoretical specific gravity of HMA mixtures (sometimes called the Rice test) is accomplished by using a vacuum to saturate the test specimen in water. The vacuum causes the water to boil and drives water into any available pore spaces that exist between coated particles. The saturated specimen is then weighed in air and in water and specific gravity values are calculated. The result is the specific gravity of the mix at zero air voids.

Recent changes to the test procedure have added more restrictions on the vacuum levels applied to the specimen and have introduced vibrating frames to produce consistent levels of agitation for each specimen. Use of these modern procedures will reduce the variability between laboratories on a given project.

Test procedures for maximum theoretical specific gravity are AASHTO T 209 or ASTM D 2041.

The test values of maximum theoretical specific gravity are used in conjunction with other tests. It is the basis of the determination of air voids when compared with laboratory specific gravities and can be a basis for compaction values when compared with cores sampled from completed pavements.

10.3.5 Bulk Specific Gravity (Laboratory Density)

Testing for bulk specific gravity is done by measuring the bulk specific gravity of laboratory compacted or cored asphalt specimens. Implementation of

the Superpave system has led to increased usage of the Superpave Gyratory Compactor (SGC) for laboratory compaction of bulk specific gravity specimens of HMA. The Marshall method and the Hveem method of compacting test specimens are also used in some jurisdictions.

The bulk specific gravity tests are typically used for two reasons: to establish a laboratory density for comparison to in-place density to determine compaction, and to calculate volumetric properties of the mixture. Regardless of the method used to prepare the test specimens (SGC, Marshall, or Hveem), the intent is to compact test specimens within a specified temperature range with a standard amount of compactive energy in the laboratory. The specimens are first weighed in air, and then weighed again while submerged in water. This testing yields specific gravity test results that can be converted into unit weight values.

The laboratory compacted density can then be used to establish a target value for compaction in the field. A typical compaction specification, for example, would require the contractor to compact the HMA in the field to 95 percent of the laboratory compacted specimens. The bulk specific gravity test results are also used to calculate air voids.

10.3.6 Air Voids

Air voids are calculated using the bulk specific gravity of compacted specimens and the maximum theoretical specific gravity. Specifications typically require the laboratory compacted air voids to be within a specified range of the air voids from the JMF (mix design). A typical range would be plus or minus one percent from the mix design value. Refer to AASHTO T 269 or ASTM D 3203 *Percent Air Voids in Compacted Dense and Open Bituminous Paving Mixtures* for calculating the mixture air voids.

10.3.7 Temperature

Temperature control is critical to many HMA operations including production, placement and compaction. The use and importance of temperature measurements are discussed in more detail in Chapter 7, "Manufacture of Hot Mix Asphalt," Chapter 8, "Hot Mix Asphalt Paving," and Chapter 9, "Hot Mix Asphalt Compaction." Temperature is generally measured (Figure 10.9) at many different points during production and placement of hot mix asphalt. It is usually measured with simple thermometers, but several adaptations to the equipment are useful, especially for quality-control. Traditionally, large-face dial thermometers, some with long stems, have been used for testing temperatures at the various stages of asphalt production and paving. Digital thermometers with replaceable probes are being used more because the costs have come down. These thermometers can be very accurate across a broad range of temperatures, so they are very versatile. Infrared thermometers are also gaining in popularity; the

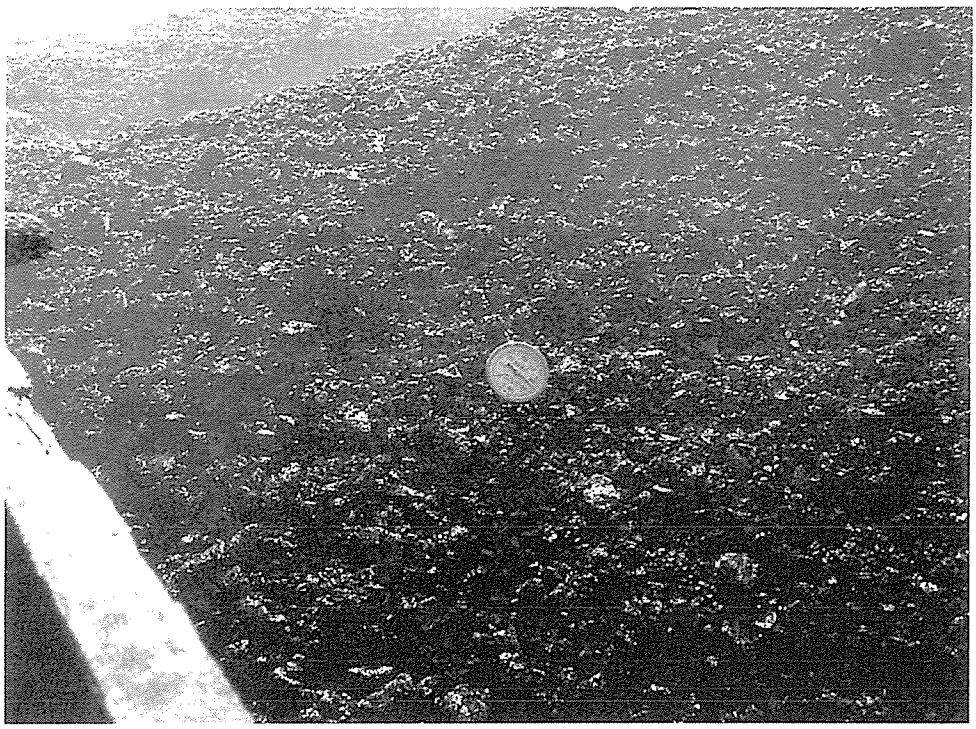


Figure 10.9 Temperature Measurements

advantage to their use is that readings are taken by simply pointing the infrared source at the object. These thermometers are useful for measurements that may otherwise pose a safety risk, such as measuring hot asphalt in the back of a haul unit. Infrared thermometers must be used with caution, however, because they actually measure surface temperatures, which can be different from the underlying material. They can be particularly inaccurate in measuring pavement mat temperatures in windy conditions or when moisture exists on the pavement surface.

10.4 Sampling and Testing Plan for Quality Control

Some form of quality-control (or process control) is provided during production of all HMA. As an example, consider a plant operator who is using a temperature gauge to measure the temperature of the heated aggregate before batching. The batch-plant operator will discard the initial aggregate that is not up to temperature, based on the results of these temperature readings. When the aggregate reaches the target temperature, the operator will proceed with preparing the first batch of HMA. This example demonstrates how the operator uses data to control and adjust the batching process. The temperature reading is a process control test.

There are two key aspects of true quality-control testing. The first is measuring specific characteristics and inspecting appropriate activities that affect the quality of the finished product. The second aspect is providing the results of those measurements and inspections in a meaningful way and within a time frame that allows adjustments to the production process. The subsequent adjustments or corrective action can thus prevent appreciable non-complying materials from being incorporated into the finished work.

The quantity of quality-control data and the level of control vary greatly from one project to another. Some project specifications require the contractor to exercise extensive control and mandate specific tests and test frequencies. Other specifications allow the contractor or the supplier to determine the level of testing based on their own judgment of what data is needed to adequately control the production operation.

10.4.1 Selection of Sampling Methods and Sample Frequency

Sampling methods and sample frequencies used for quality-control are sometimes established by the project specifications and sometimes selected by the contractor/supplier. Many pavement owners or agencies dictate sampling methods and give minimum sample frequencies. The contractor, of course, is free to perform any additional testing he deems appropriate.

Methods and locations of sampling can have significant influence on the outcome of the parameters being measured for quality-control. Sampling, therefore, is an important aspect of the overall quality-control program.

The methods and frequencies selected for sampling are not always the same for quality-control as they are for acceptance. An example is the sampling methods used to determine the aggregate gradation. Some agencies use cold-feed belt sampling methods for testing aggregate gradation for

quality-control purposes. These samples are taken at the asphalt mixing plant and can provide test results within hours of sampling. The results, therefore, do not accommodate "real time" adjustments but do provide opportunities for production to be adjusted at many times during a given day. In contrast to these quality-control sampling methods, the acceptance samples for gradation are often taken from the paving laydown process, and gradations are run after the asphalt binder has been extracted from the HMA samples. The results may be more representative of the mix that was delivered to the project, but they may not be received until days after placement. The amount of time needed to get results is an important factor in the selection of sampling methods for HMA production.

10.4.2 Selection of Testing Methods

In the past, control of HMA production was almost exclusively driven by asphalt content and gradation. As production capabilities become more refined, and as the technical understanding of the factors affecting longevity of asphalt pavements progresses, there is an increased emphasis on controlling air voids and other volumetric properties of the mix. Many agencies have now added mixture volumetric tests to their requirements for both quality-control and acceptance for production of HMA.

Based on NCHRP Synthesis report 346 "State Construction Quality Assurance Programs," most state highway agencies in the U.S. select the sampling and testing methods that are used for quality-control on their projects. These testing requirements are typically identified in the project specifications. Many agencies include quality-control line items in bids to reimburse contractors for costs associated with providing the tests. The synthesis report identified the following parameters that are normally required for quality-control testing programs.

- Asphalt content
- Aggregate gradation
- Compaction
- Volumetric properties (air voids, VMA, VFA)
- Ride quality
- Aggregate characteristics (CAA, FAA, F&E, SE, etc.)
- Pavement thickness

Many of the test values will be compared with the JMF. When that is the case, the same tests should be used for quality-control as were used in development of the job-mix formula. It would not be advisable, for example, to use Marshall density values to evaluate air voids on mixes that were designed using the SGC. Portable SGC equipment is available and being utilized for quality-control and acceptance on many projects. For more information on the Superpave system or the SGC, refer to the Asphalt Institute's manual SP-2 "Superpave Mix Design" and Chapter 4, Hot Mix Asphalt (HMA) Design.

For tests that are not compared with the JMF, the test methods selected for quality-control purposes may be different from the test methods used for

acceptance of the same parameter. Quality-control testing for compaction of asphalt pavements, for example, quite often uses nuclear density test methods, whereas the acceptance testing for compaction may be performed using bulk specific gravity test methods on cores cut from the finished pavement. Test results from cores come too late to be used for adjusting the compaction operation because the asphalt will have already cooled. Nuclear density tests can be taken while the HMA is still hot and the compaction operation is ongoing. Thus, nuclear density test results can be an important part of the overall quality-control program for HMA compaction.

Since quality-control methods must emphasize timeliness and convenience, the results can be less accurate and impart more variation in the data. If the QC test method is different than the acceptance method, a minor shift in the data should be anticipated. It would be unreasonable, for example, to stop compaction operations when a reading of 95 percent compaction was read on a nuclear density gauge, if the minimum required compaction was 95 percent using core densities. A higher target could be established to account for the likely variation and the possible shift of compaction values from the nuclear density method to the core density method. It is also a prudent quality-control measure to correlate quality-control test methods and laboratories to the acceptance methods performed by the acceptance laboratory. This can be accomplished with several "split" samples from a test strip or at the beginning of HMA production.

If a nuclear gauge is being used to control density and if an accurate correlation to core densities has not been made, a "maximum density" method can be used to overcome the variation from nuclear density test results and core density test results. For this maximum density method, nuclear density readings are taken in specifically marked locations between roller passes so that the relative increases in compaction can be tracked throughout the compaction operation. When the pavement reaches the highest achievable density (that is, other rolling patterns yield lower density), then the pavement has achieved a relative maximum density. Compaction should continue with the same pattern until this relative maximum is achieved in other locations for that day's placement as well. Establishing rolling patterns is a trial-and-error process. It is important to recognize that the number of vibratory passes in the breakdown rolling will typically have a greater effect on the final compaction values than the number of passes with a finish roller. Therefore, rolling patterns with varying initial breakdown passes should be included in the trial-and-error process. A variety of rolling patterns may need to be tried before settling on the final pattern. For more information on compaction of hot mix asphalt, refer to Chapter 9, "Hot Mix Asphalt Compaction."

10.4.3 Written Quality-Control Plan

Most agencies that specify quality-control sampling and testing on their projects also require the contractor to submit a quality-control plan before beginning paving operations. The requirements for these plans can be

extensive and usually involve more than just sampling and testing plans. They will generally require documentation verifying compliance with certain project requirements related to the quality-control program. The following items are usually required as part of written quality-control plans for HMA production based on the NCHRP Synthesis Report:

- Responsibility matrix showing who is responsible for reviewing quality-control results and directing any corrective action
- Sampling and testing plan identifying specific tests and frequencies to be used for the project
- Documentation demonstrating that the plant meets the required qualifications, certifications, and inspection frequencies
- Documentation demonstrating that the laboratory that will provide the quality-control testing meets the required qualifications and accreditations
- Documentation demonstrating that the technicians who will provide the sampling and testing meet the required qualifications and certifications
- Example report forms, including control charts that summarize certain key data on plots that can rapidly show trends that need attention
- Written procedures describing criteria that will be used to identify "out of control" production, what action steps will be taken when materials fall outside of those criteria, and what the timing/deadlines of these actions will be

An example sampling and testing plan that would be a part of the quality-control plan for HMA production is presented in Table 10.6. Both the ASTM and the AASHTO methods are listed in Table 10.6 for the convenience of the reader. The AASHTO and ASTM methods are not interchangeable because they are not always the same. Specifications should designate the method to be followed.

The timing of sampling and testing as well as the turnaround of the test results is critical for the quality-control plan to effectively control the quality of materials produced. If a belt sample is taken at the beginning of production and tested for gradation, the test could be completed in a matter of hours and reported back to the plant manager. If those test results are not provided in a timely way to the plant manager—or other person in charge of the plant—then the data may contribute nothing in controlling the plant process. Timeliness in communication of test data is essential, especially if the quality-control laboratory is not directly employed by the HMA producer. Good quality-control plans must address these lines of communication as well as the timing and responsibilities of the various players. When several different companies are all involved in the quality-control program, formal and informal lines of communication must be established in order to offset potential barriers created by the various contractual responsibilities of each party.

After all the documentation for the project has been gathered, the written quality-control plan is generally submitted to the owner for review

Table 10.6
Example Quality Control Sampling and Testing Plan for HMA

Test Parameter	Sample Point/ Method	Test Method, AASHTO	Test Method, ASTM	Sample Frequency	Estimated Quantity	No. of Tests
Aggregate Gradation	Cold Feed Belt, ASTM D 75	T 27, T 11	C 136, C 117	1000 tons (907 tons)	28,000 tons (25,400 tons)	28
Fine Aggregate Angularity	Cold Feed Belt, ASTM D 75	T 304	C 1252	Each day	7 days	7
Asphalt Content	Plant or Truck, ASTM D 140	T 308	D 6307	1000 tons (907 tons)	28,000 tons (25,400 tons)	28
Laboratory Air Voids	Plant or Truck, ASTM D 140	T 312, T 166, T 209, T 269	D 4013, D 2726, D 2041, D 3203	1000 tons (907 tons)	28,000 tons (25,400 tons)	28
Compaction	Pavement Surface		D 2950	1000 feet (305 meters) for each paving lane	53,000 feet (16,154 meters)	53
Pavement Thickness	Pavement Surface, by coring		D 3549	1000 feet (305 meters) per paving lane	53,000 feet (16,154 meters)	53
Ride Quality, Smoothness	Pavement Surface	10-foot (3 meter) straightedge, (see spec.)	10-foot (3 meter) straightedge, (see spec.)	50 feet (15 meters)	53,000 feet (16,154 meters)	1060

and approval. The plan is usually submitted with other project submittals related to the HMA production like the HMA mix designs for all the HMA mixtures to be used on the project, aggregate source qualification testing, certifications for commercially provided materials that will be incorporated into the HMA, designation of what plant or plants will be used for production, and paving plans. After the review and approval process is completed, the contractor is generally authorized to begin production, unless the project specifications call for production of a test strip.

10.4.4 Test Strip

Every project involves a unique set of circumstances and challenges relative to producing a quality pavement. Even the best pre-paving planning leaves the contractor with methods and materials that work in theory but have not been proven in actual production. A job-mix formula, for

example, is established based on mix design analyses from samples that have been prepared and mixed in the laboratory. The results of these tests do not exactly match the results of similar materials that are produced in an industrial asphalt mixing plant. Placement and compaction operations usually have to be adjusted from one project to another because of differences in the environment and the equipment used. Compacting asphalt on compacted subgrade, for example, may require different rolling patterns from that for an overlay of an existing HMA roadway. Test strips to validate the means, methods, and materials to be used under the specific project conditions are becoming widely used by owner/agencies and paving contractors.

Test strips or "test sections" are constructed before beginning regular production paving for a given asphalt mixture. They are typically considered part of the regular pavement if the test results comply with project specifications. They are comprised of shorter lengths of pavement (typically 500 to 1,000 feet or 154 to 308 meters) and should be constructed with the same equipment and materials that will be used for the rest of the project. Projects with cold joint density specifications will often require a cold joint to be constructed as part of the test strip. Two or three days of time may be required to evaluate the test strip data before beginning regular production paving. Testing for test strips is typically the same as that required for quality-control and acceptance for the rest of the project. Comparing test results from different laboratories and confirming the details of the methods used can be a side benefit to the test strip process. This is a good time to verify that any variations in laboratory values are a function of normal deviation in testing and differing methods and are not because incorrect or different procedures or equipment are being used by one of the laboratories.

If the test results of the test strip indicate the job-mix formula is suitable for producing complying materials, and the appropriate spreading, placing, and compaction can be achieved, then production paving can proceed. If the results of the test strip do not meet project specifications, then adjustments can be made and a new test strip produced.

10.5 Plant Production Quality-Control Data Analysis

Quality control of plant production involves producing, analyzing, and reporting a tremendous amount of data. The organization of this data is generally established in the quality-control plan. There are three key areas for which the data must be properly analyzed, documented, and understood. The first of these is in verifying the job-mix formula. Because the formula is developed in the laboratory and made from samples taken weeks before production begins, it is always prudent to verify that the job-mix formula will produce a quality mix and that the parameters established in the mix design can be achieved. The second key area is organizing and documenting the daily quality-control data. This is necessary so that the data is meaningful to the users and readily available if problems occur or if adjustments need to be made. The third key area is in adjusting the production process. A good understanding of the test results and reasons for the data are the foundation of the good judgment necessary to make appropriate adjustments. More detailed descriptions of these three key areas of quality-control data analysis are presented below.

10.5.1 Job-Mix Formula Verification

Field verification of the HMA mix design is the first step in implementing the overall quality-control plan. It involves testing and analyzing the field-produced mixture to verify that the criteria established by the specifications are being met. Verification is necessary at the beginning of production of each mixture or job mix formula to determine what differences exist between design and production and if adjustments need to be made.

HMA mixture designs are developed in the laboratory, usually with aggregates sampled from the project and before production begins. Testing and analysis of the anticipated performance of that mix is generally accomplished as part of the design.

It has been documented in many cases that even though a mix is designed for a certain percent of air voids in the laboratory, a plant-mixed sample containing the proper asphalt content, aggregate gradation, and compaction by the same laboratory method may show different volumetric properties. In their study of field testing and field laboratory test results titled FHWA Demonstration Project 74, "Field Management of Asphalt Mixes," FHWA found substantial differences between the laboratory mix design and the volumetric properties of the field-produced mixture. Based on the study, these changes tend to result, on average, in slightly lower air void and VMA values than the targets established by the laboratory-prepared JMFs.

Significant equipment and material differences exist between the small-scale operation of the laboratory mixing bowl and an asphalt mixing facility, which could lead to the mixture property changes noted. Many causes for volumetric changes in field-produced mixture have been proposed:

- Moisture removed more effectively in the laboratory than in the field
- An excessive amount of baghouse fines returned to the mixing process
- Highly absorptive aggregates used. (Field absorption may differ from laboratory absorption.)
- Poor sampling used to obtain aggregate samples for the laboratory design
- Nonstandard mix design procedures used to develop the JMF
- Material sources (location and/or strata) changed between the mix design and field production
- Aggregate degradation and shape changes during plant mixing
- Non-uniform stockpiling of aggregates

The changes that occur in the field-produced mixture are likely due to a combination of some or all of the listed causes in any particular case. Attention to detail in material production and mix design development can reduce the potential effects of many of the factors shown.

Some form of verification that the proposed job-mix formula will produce materials meeting the project specifications will occur on every project. Some agencies require that verifications be performed by the agency laboratory or by a laboratory different from the one that did the mix design. Other agencies will utilize test strips to validate the job-mix formula. Additional discussion of test strips can be found in Section 10.4.4. Some agencies allow the contractor to make minor changes in the job-mix formula without a new mix design if the changes are made to improve the volumetric properties of the mix. Improvements to volumetric properties in such cases generally mean getting the laboratory air voids closer to the target value.

If there are no specified means to verify the job-mix formula, then it will be done during the initial paving. This creates a higher risk that non-complying or marginal HMA will be produced for a longer period of time before adjustments are made.

10.5.2 Mixture Control

Quality-control test data must be quickly reported and analyzed so that it can be used to improve the consistency and overall quality of the HMA product. The goal is to understand the data and to evaluate whether any adjustments to the plant process or materials processes are warranted. It is necessary to keep thorough records so the quality-control manager can understand the overall production and observe trends that may need further action. Good mixture control is accomplished by first organizing the data in a meaningful way. Control charts are often used for this. Then the data must be analyzed and understood. Finally, appropriate adjustments to the production operations need to be made.

Control Charts

One way that quality-control data is presented is on control charts (Figure 10.10). Control charts are graphed plots of quality-control data over time. The specification limits may also be plotted. Control charts are useful because they provide a quick visual reference of how close to the specification limits the test data are. They can also be invaluable in spotting trends in the data.

Data Analysis

Some data collected for quality-control is straightforward, and corrective action is relatively automatic after receiving the information. An example of this would be measurements of temperature at the batch plant. If the target temperature for mixture leaving the plant is 320°F (160°C), and the quality-control check determines that the temperature is 300°F (149°C), the plant operator can begin adjustments immediately to increase the mixing temperature to the target.

Other data collected for quality-control requires careful analysis and consideration before corrective action is taken. An example of this would be measurements of laboratory air voids in the mix. If the target air voids are 4 percent (assume the specifications limit is ±1 percent) and if the quality-control test indicates the mixture is at 2.9 percent, there are several things the plant operator and the quality-control managers would need to consider

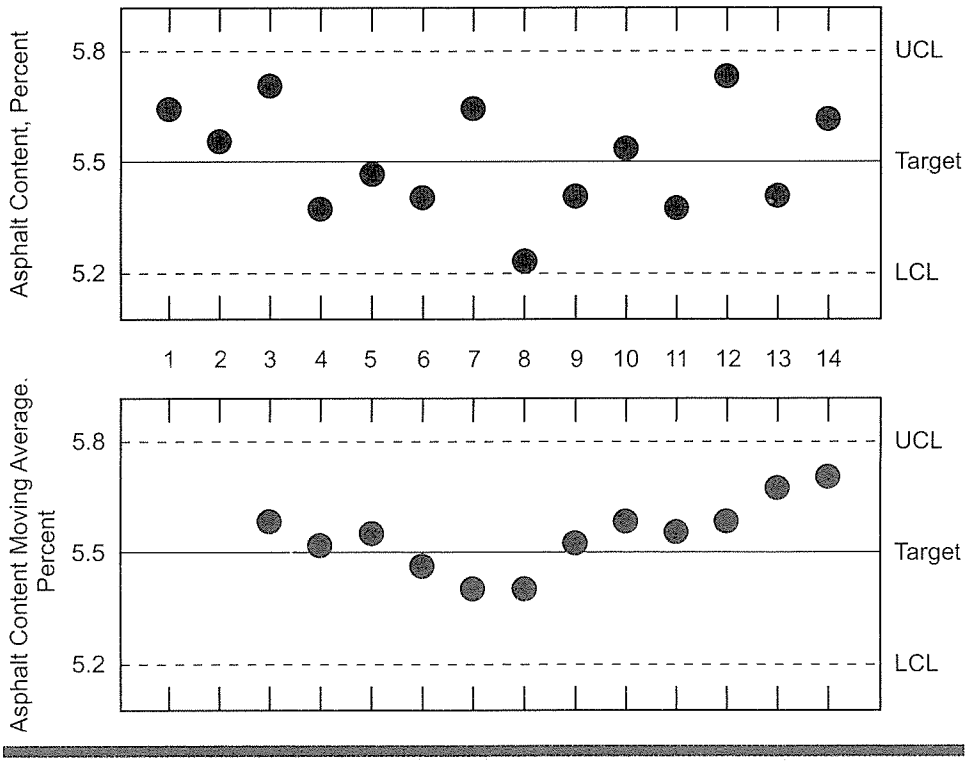


Figure 10.10 Control Chart Example

before establishing if any corrective action was warranted. Perhaps the first thing to investigate is the reliability of the sampling and testing for the particular result. It is common to immediately take additional samples and get retesting under way while the further investigation is ongoing. Other related test results should also be evaluated to see what other parameters have changed. Key test parameters related to the air voids would be asphalt content, percent passing the No. 200 (0.075 millimeter) screen, aggregate gradation (especially on screens between the No. 8 (2.36 millimeters) and No. 40 (0.425 millimeter) sizes), and fine aggregate angularity. The investigation would also question whether any adjustments were made to other plant processes such as percentages of aggregate stockpiles fed into the plant or sources of asphalt cement. Any one of these things, or a combination of some of them, might cause a change in air voids. An understanding of the cause of the initial change in air voids is needed to make a good decision about what adjustments would be appropriate.

Regardless of the actions taken, it is imperative that complete and accurate records of the production and quality-control activities be kept while the work is ongoing. This information and these records will be needed in order to investigate difficult mixture issues as they come up. The investigation can be rapid and accurate if proper records are kept, but it will be long and drawn out if the quality-control manager and the plant operators have to rely on the recollection of a variety of people to attempt to piece together specific information about what happened.

Quality-control plans usually describe the documentation that will be provided during the project. The use of computer spreadsheets, e-mail, and the Internet mean these reports can contain large amounts of data, and the information can be distributed to the appropriate parties in a timely way. Because of the tremendous amount of data generated for quality-control of HMA production, summaries of the daily average values can be useful in analyzing and understanding how the materials are changing over time and the relative level of control that is being accomplished. An example of this type of summary is presented in Figure 10.11.

It is impossible to plot all the relationships in the quality-control data that may become useful. Storage of quality-control data in spreadsheet applications can be helpful so that specific plots can be rapidly developed while investigating noncompliant items. An example of this is shown in Figure 10.12. Consider a quality-control manager investigating high air voids with the data presented in the quality-control summary (Figure 10.11). A plot of air voids versus minus No. 200's reveals a strong relationship between the two parameters. As the minus No. 200's decreased, the air voids decrease as one would expect. The plot verifies that this is in fact what is happening and suggests that better control of the minus No. 200's would affect better control of the air voids. Further observation of the data reveals that the mixture is essentially achieving the design air voids if the mix design minus No. 200's is achieved. In this example, changes in the job-mix formula would not contribute to correcting the high air voids.

Figure 10.11 Quality Control Summary Example

Plant Production Quality Control Summary

Paving Date		Lot	Sub lot	Station		Lane	Gradation, % Passing			Asphalt Content	Gyratory			
				from	to		#4 Sieve	#30 Sieve	#200 Sieve		Bulk Sp. Gr.	TMD	Unit Wt.	Air Voids
				Action Limit			51-63	15-23	2.6-5.4	4.6-5.4				
				Suspension Limit			49-65	13-25	2.0-6.0	4.5-5.5				2.6-5.4
9/9/06	Sat	1	1	120+30	134+30	L1	58	17	4.3	5.07	2.356	2.444	146.6	4.2
Taxiway Alpha South			2			L2	60	18	2.5	5.10	2.350	2.462	146.3	5.5
1st Lift			3			L3	61	15	5.0	5.08	2.345	2.448	146.0	4.2
			4			L4	57	16	4.5	5.00	2.352	2.444	146.4	4.3
9/12/06	Tue	2	1	14+30	120+30	1	62	16	3.7	5.02	2.325	2.439	144.7	4.5
Shoulder South			2			1	62	15	2.9	5.04	2.336	2.444	145.4	5.1
			3			1	58	18	3.1	5.15	2.345	2.448	146.0	5.0
			4			1	59	14	3.8	5.20	2.316	2.429	144.1	4.7
9/13/06	Wed	3	1	1+90	14+30	L1	64	12	3.0	4.88	2.334	2.451	145.2	5.4
Taxiway Alpha North			2			L2	63	19	3.5	5.04	2.332	2.543	145.2	5.0
1st Lift			3			L3	61	18	3.5	4.92	2.336	2.457	145.4	5.0
			4			L4	59	13	3.4	4.88	2.316	2.429	144.1	5.2
9/14/06	Thu	4	1	14+30	120+30	1	55	17	4.0	5.00	2.364	2.448	147.1	4.8
Shoulder North			2			1	54	16	4.5	4.78	2.363	2.442	147.1	4.5
			3			1	53	13	4.3	4.65	2.334	2.456	145.3	5.0
			4			1	54	15	5.0	4.85	2.354	2.449	146.5	4.4

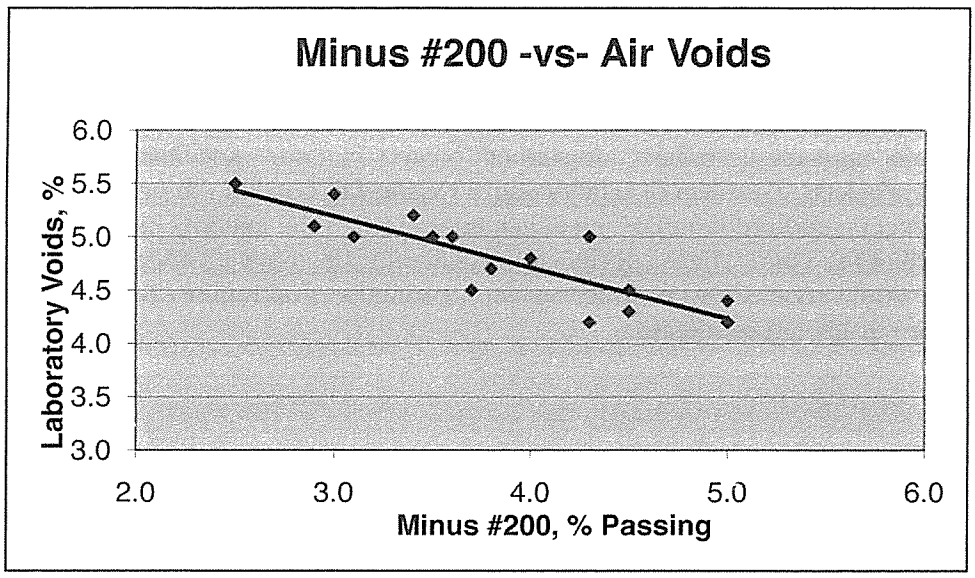


Figure 10.12 Air Voids Investigation Example

10.5.3 Plant Adjustments

The purpose of quality-control testing for HMA production is to provide plant managers and operators with valuable information they need to control the plant. If the data show that the produced material is deviating from the job-mix formula, the plant personnel should decide what adjustments are needed to bring the material closer to the job-mix formula.

It is important to realize that all HMA plants (in this discussion, aggregate crushing and production equipment are considered part of the plant) will produce material that has a given amount of variation. Some plants will produce materials that are more consistent than others. Understanding this is important because adjusting the production will not correct normal plant variations.

As an example of this, consider a plant that is operating under an allowable tolerance for the percent passing the No. 200 screen of 3 percent to 7 percent. Assume the average result from quality-control tests is 6 percent passing the No. 200 (0.075 millimeter) and the QC test results range from 4.5 percent to 7.5 percent. No adjustments have been made to the plant up to this point. In this example, the job-mix formula for minus No. 200 (0.075 millimeter) was 5 percent and the average QC results were 6 percent. Based on this information, the plant operator should be considering an adjustment that could slightly reduce the minus No. 200, preferably by 1 percent. All the other quality-control data, of course, would have to be reviewed before any adjustments are made to be sure that it would not be detrimental to other quality-control test parameters.

Assume in the above example, that the first quality-control test result received was the 4.5 percent minus No. 200 (0.075 millimeter). An immediate

adjustment to increase the No. 200 (0.075 millimeter), in an attempt to get them closer to the job-mix formula, would be a mistake in this instance. It would be a mistake because it would ignore the normal variation of plant production.

Some agencies establish "adjustment" criteria in their specifications. For those projects, if results of quality-control testing fall outside the criteria, the producer is required to make an adjustment to bring the material closer to the middle of the band. These specifications generally also have a wider criterion that would require the producer to stop production and adjust the mixture before restarting production.

10.6 Placement Quality-Control Data Analysis

The primary quality factors for placement of HMA are compaction, thickness, and smoothness. Placement quality-control, therefore, focuses on these same factors. These three factors are discussed below. Temperature is also discussed because it plays such a vital role in controlling and achieving compaction.

Similar to quality-control data for hot mix production at the plant, it can be very useful to summarize the placement quality-control data to get a better overall picture of what is happening during placement and compaction. The two key factors measured—smoothness and compaction—are critical to the performance of an asphalt pavement, so understanding and controlling these aspects of the construction operation are very important. Tables 10.7A and 10.7B (metric) show examples of a quality-control data summary for HMA placement. For complete quality-control documentation, summaries should be compiled and given to the quality-control manager on a regular basis. For interpretation and analysis of the data, it is useful to get an overall understanding of what is happening on the project and then to evaluate the details of particular issues more specifically.

10.6.1 Temperature

Temperature is perhaps the most critical variable in HMA compaction. The temperature of the mix is, therefore, very important to those responsible for compaction. One difficulty that the compaction crews have is that the HMA goes through so many other operations before it is ready to be compacted. The temperature is affected by each operation, and the compaction equipment operators and foremen have little influence on them. Good quality-control, which includes temperature readings at all phases of production and delivery, will help provide HMA at the appropriate compaction temperatures to the rollers. Some quality-control specifications require temperature readings to be recorded at many points along the way and reported each day. Specifications for a project using a batch plant, for example, may require that temperature readings be recorded for the asphalt cement prior to mixing, the aggregate after heating, the mixture immediately after mixing, and at delivery to the paving machine. With this kind of data stored for each day's production, it would be easy to determine the root cause of temperature problems if they occurred on a particular day. More importantly, it would be easy to establish a corrective action plan to get the temperatures where they need to be for proper compaction.

Table 10.7A
Quality-Control Placement Data Summary Example

Placement Quality-Control Data Summary

Sample Description				Marshall	Compaction			Temperature (°F)			
Paving Date	Day	Lot	Sublot	Voids	Sta	Offset	Percent of Max.	AC Tank	Agg. Dryer	Plant	Site
		Target Values		*4*			*93 min.*			*310*	*290*
10/25/06	Wed		1	3.7	82+28	7	96.8	300	330	325	290
Deadhorse Pass			2	4.6	85+88	4	94.2	300	330	308	300
First Lift		1	3	4.2	88+88	7	94.6	300	330	325	319
			4	3.8	96+63	2	92.4	300	315	315	311
		Average		**4.1**			**94.5**	**300**	**326**	**318**	**305**
10/26/06	Thurs		1	5.0	98+60	2	93.1	300	290	300	290
Deadhorse Pass			2	5.1	102+19	6	91.4	300	310	305	292
First Lift		2	3	5.1	109+64	2	93.2	300	280	290	280
			4	4.9	113+27	7	91.5	300	300	300	285
		Average		**5.0**			**92.3**	**300**	**295**	**299**	**287**
10/27/06	Fri		1	3.4	116+68	5	95.0	320	320	315	300
Deadhorse Pass			2	3.2	121+77	6	94.7	320	320	315	295
First Lift		3	3	5.0	125+50	7	94.0	320	320	315	305
			4	3.9	130+91	6	92.8	320	310	310	300
		Average		**3.9**			**94.1**	**320**	**318**	**314**	**300**

Table 10.7B
Quality-Control Placement Data Summary Example (Metric)

Placement Quality-Control Data Summary (Metric)

Sample Description				Marshall	Compaction			Temperature (°C)			
Paving Date	Day	Lot	Sublot	Voids	Sta	Offset	Percent of Max.	AC Tank	Agg. Dryer	Plant	Site
		Target Values		*4*			*93 min.*				
10/25/06	Wed	1	1	3.7	2+507	2.1	96.8	149	165	163	143
Deadhorse Pass			2	4.6	2+617	1.2	94.2	149	165	153	149
First Lift			3	4.2	2+709	2.1	94.6	149	165	162	159
			4	3.8	2+854	0.6	92.4	149	157	157	155
		Average		**4.1**			**94.5**	**149**	**163**	**158**	**151**
10/26/06	Thurs	2	1	5.0	3+005	0.6	93.1	149	143	149	143
Deadhorse Pass			2	5.1	3+115	1.8	91.4	149	154	151	144
First Lift			3	5.1	3+342	0.6	93.2	149	280	143	137
			4	4.9	3+452	2.1	91.5	149	149	149	140
		Average		**5.0**			**92.3**	**149**	**146**	**148**	**141**
10/27/06	Fri	3	1	3.4	3+556	1.5	95.0	160	160	157	149
Deadhorse Pass			2	3.2	3+711	1.8	94.7	160	160	157	146
First Lift			3	5.0	3+825	2.1	94.0	160	160	157	152
			4	3.9	3+990	1.8	92.8	160	154	154	149
		Average		**3.9**			**94.1**	**160**	**159**	**157**	**149**

10.6.2 Compaction Control Methods

The goal of compacting a hot mix asphalt pavement is to achieve an optimum air void content and provide a smooth, uniform surface. The resulting in-place air void content of the HMA is in many ways the single most important factor that affects performance of the mixture throughout the life of the pavement. The activities involved in the proper design, production, placement, and compaction of the asphalt mixture are combined to achieve the in-place density of the HMA pavement and to ultimately determine whether the pavement will perform as expected.

The key points here are that controlling compaction is very important and that compaction is affected by many factors other than those specifically measured during placement of the mixture. This means that low compaction problems on a given project may not be resolved simply by adjusting the rolling pattern or the temperatures. The quality-control manager and project superintendent may need to analyze all the quality-control data, including that from the mixture production, to address compaction problems.

Of course, in addressing a compaction problem, the paving crew should first review their own operations and procedures for the placement and compaction operation, paying particular attention to temperatures, disciplined adherence to good rolling technique, and consistent and tested rolling patterns. The compactability of the asphalt mixture provided should also be evaluated.

A variety of methods are available to control the quality of the compaction for HMA, and those methods will be discussed in more detail below. These methods are usually driven by specification, however, so the quality-control will be based on the specified methods for acceptance of compaction. The quality-control efforts should be centered on establishing and documenting compliance with the actual specifications.

In-Place Density/Compaction Control Measurements

A common control method for compaction is to simply measure the in-place density. These measured densities will be compared with a laboratory density (the specified reference density), and a percentage of the laboratory density will be calculated. Nuclear density measurements are the most common form of quality-control density testing. Non-nuclear devices using electromagnetic waves to measure the dielectric constant in the pavement materials have been developed and are gaining popularity for quality-control density testing. Information on comparisons of nuclear and these non-nuclear devices can be found in the reference list. In some instances, quality-control testing for density is done using cores; however, the use of cores imparts a delay in the results. Cores should always be used, however, to correlate and/or calibrate the quality-control test methods selected, especially if cores are to be used for acceptance.

In-place density tests are used to measure compaction. As an example, consider a project that specifies compaction to be at least 96 percent of Marshall specific gravity, and the Marshall specific gravity is 2.324. This specific gravity value can be converted to density by multiplying it by the unit weight of water (62.4 lb/ft^3) (1000 kg/m^3) which yields 145.0 lb/ft^3) (2324 kg/m^3). If the quality-control testing yields an in-place density of 139.2 lb/ft^3) (2228 kg/m^3) at a given location, then compaction is calculated as follows:

$$\text{Compaction} = \left(\frac{139.2}{145.0}\right) \times 100 \text{ or } \left(\frac{2228}{2324}\right) \times 100 = 96.0 \text{ percent}$$

This test result meets the minimum compaction, but additional effort would need to be applied to account for variations in testing and the construction operations. It may be more appropriate to target 97–98 percent, or maximum achievable density, for quality-control purposes on projects with a 96 percent minimum requirement.

Rolling Patterns

Most paving contractors will create defined "rolling patterns" for a particular placement on the first day of paving. Nuclear density test data can play a key role in developing these rolling patterns. A rolling pattern is defined as the number and sequence of successive passes of compaction equipment over an asphalt pavement. The rolling pattern is generally adjusted until the desired compaction is achieved. It is a trial and error process. Details of compaction principles and rolling procedures can be found in Chapter 9, "Hot Mix Asphalt Compaction."

In order to provide meaningful information to the quality-control manager, the quality-control density testing must be done at many points in the rolling operation. Rolling patterns are established on the first day of paving and ideally done during construction of a test strip. The test strip will provide calibration of the nuclear gauge with cores cut from the pavement, and the rolling pattern may need to be adjusted. The following example describes how the quality-control testing is used in the development of a compaction control-rolling pattern.

Rolling Pattern Example

The equipment that will be used for the rolling pattern must be the same as will be used during construction. For this example, the compaction equipment consists of a double-drum vibratory compactor for the breakdown rolling, a heavy pneumatic compactor for the intermediate rolling, and a static steel-wheel compactor for the finish rolling.

During breakdown rolling, the operator may make a vibratory pass forward and back for the first coverage. After that coverage, the quality-control technician takes a nuclear density test and marks the location of the test on the pavement. This is done by drawing a rectangle outlining the bottom of the gauge on the pavement so the gauge can be returned to the same location and orientation for subsequent tests. It is good practice to mark and test multiple locations so that average density changes can be

used. The location for testing should be in an area where there is no turning, stopping, starting, or anything else that will affect the density readings. In the example, the project acceptance is based on a minimum of 93 percent of maximum theoretical density. Assume that the first density test result is 90.5 percent compaction. Next, the roller operator makes another coverage, this time consisting of a static pass and a vibratory pass. After that coverage, the technician places the gauge back in the exact location and takes a second reading. This time the technician records a compaction value of 91.5 percent. At this point, the paving superintendent must decide whether to go to the next phase of compaction or to attempt to gain additional density with the breakdown roller. Assume the superintendent decides to make another pass with the breakdown roller, and the next density result is 92.0 percent.

Now, the breakdown roller moves on and the intermediate roller continues the compaction process. During intermediate rolling, it may not be necessary to test after each coverage, but a couple of readings to show progression of density should be made. In this example, assume that after three passes with the intermediate roller, the compaction reading is 93.0 percent and the superintendent decides to move on to the finish roller.

The finish roller will work to smooth the surface of the pavement and the joints but will not necessarily gain additional compaction. This depends on a number of factors such as the temperature of the mat and the compaction achieved before finish rolling. See Chapter 9, "Hot Mix Asphalt Compaction" for more information on compaction methods. For this example, assume that after two coverages with the static roller, the nuclear density gauge reads 93.5 percent, then after another coverage, there is no gain in density. At this point, depending on variables such as temperature of the mat, additional compaction from the finish roller is not likely, and the finish roller moves on to the next portion of pavement. There is now a rolling pattern and measured compaction results for that pattern. The same rolling pattern is expected to give similar density results if used again under the same conditions.

The paving superintendent now has one rolling pattern and a result of 93.5 percent on a project that specifies a minimum of 93 percent. A rolling pattern achieving only 0.5 percent above the minimum is probably not very comforting to a superintendent who will have many daily variations because of weather, mix temperature, mix properties, subbase preparation, and so on. Perhaps the superintendent will try another rolling pattern on a new portion of the pavement and use different sequencing of compaction. A new pattern may include a different number of passes, a different amplitude on vibratory compactors, a different tire pressure, a ballast weight change on pneumatic rollers, heavier compactors, insertion of another vibratory roller into the intermediate compaction process, and so on. Numerous options are available for compaction rolling patterns.

With regard to the quality-control test results, it is important to remember that nuclear density gauges can have a large bias and wide variation from the actual density of the pavement. Good quality-control plans improve the accuracy of these instruments by calibrating the gauge with cores from the project. The calibrations are sensitive to lift thickness and mix design changes. Any time

a new mix or a different mat thickness is used, new calibrations for nuclear gauges and new rolling patterns should be established. Before obtaining these calibrations, direct density readings without calculating a percent compaction can be used. In this way, the development of rolling patterns must rely on relative density changes instead of measured values of percent compaction.

10.6.3 Mixture Compactability

Some mixtures are easier to compact than others. It can be useful when controlling the paving operation to understand the effect of the mixture and the variables that affect compactability.

Experience says that coarse-graded mixtures with low fines tend to require significant effort to compact. Success with such mixtures can be achieved with excellent compaction methods and good quality-control measures.

Two other mixture parameters that should be of particular interest to the paving crew or paving subcontractor are air voids and lift thickness. The following sections describe these two mixture properties and how they can affect compaction. These two items should be carefully planned and monitored with respect to their impact on the compactability of HMA.

Laboratory Air Voids

If a mixture is produced with laboratory compacted air voids that are higher than the target, additional compactive effort in the field will be required to achieve density. Suppose, for example, that a project has an established rolling pattern that achieves about 94 percent compaction when the mixture is produced at 4 percent laboratory voids. If the laboratory air voids of the mixture suddenly increased by 1.5 percent (to 5.5 percent), then the same rolling pattern in the field would yield something on the order of 1.5 percent less (about 92 to 93 percent).

When changes in laboratory air voids create a need for additional compactive effort, the establishment of a new or revised rolling pattern may be needed. Even if no air void limits are specified for production, the laboratory compacted voids must be controlled to produce a compactable mix.

Compaction problems can occur as well if the air voids are low, especially in projects with statistical acceptance requirements. In those cases, fluctuations in air voids will cause fluctuations in compaction and can affect pay factors that include standard deviations. Good consistency in laboratory air voids of the produced mix is essential for maintaining good control of compaction.

Lift Thickness

Lift thicknesses must be well planned, and compactability should be the driving factor in that decision. If the desired lift thickness is less than the recommended values, then a mixture with a smaller nominal maximum aggregate size should be used so that proper compaction can be achieved. See Chapter 9 for recommended lift thicknesses based on nominal maximum aggregate size.

Lifts that exceed the minimum lift thicknesses are especially important with very coarse-graded Superpave mixtures. See the Asphalt Institute's SP-2, *Superpave Mix Design* for an explanation of aggregate gradations and the restricted zone. These coarse-graded Superpave mixtures require thicker lifts than fine-graded mixtures, and they can tend to require more compaction energy than finer mixtures. In order to control and achieve required compaction, proper lift thicknesses must be maintained.

It is also important to realize that adjustments to compaction operations are required any time the lift thickness changes. New lift thicknesses necessitate revisions to nuclear density calibrations and rolling patterns, as well as reconsideration of timing and temperature.

10.6.4 Thickness and Spread Rates

Quality-control testing for pavement thickness is accomplished using depth gauges (Figure 10.13) that are pushed into the HMA after the paver has placed the hot mix but before the compaction equipment has rolled it down. For most dense-graded mixes, a rule of thumb is that the thickness of the uncompacted hot mix should be 1.25 times the final thickness. The actual decrease in thickness could be more accurately determined for a given mix design using core thicknesses measured and compared with the initial depth gauge thicknesses.

Thickness is also measured by cutting cores from the completed pavement surface. Core thicknesses are considered the most accurate method, but because they are taken after the work is complete, they do not lend themselves well to quality-control purposes. These values could be used to adjust thicknesses for subsequent paving days, but the paving equipment is usually controlled with thickness readings behind the paver with a depth gauge.

In measuring and adjusting lift thicknesses, it is important to consider that paver screeds are self-leveling. They are designed to fill in the low spots and cut down the high spots, making it impossible to lay a uniform lift thickness on an uneven surface. If the depth checks were exactly equal all of the time, the new surface would have the same rough profile as the underlying pavement or base. In reality, many quality-control "depth checks" will be made behind the paver, but these checks should lead to very few adjustments to the paving machine. Consideration of this natural variation in lift thickness, especially for overlays on rough pavements, is necessary for evaluating core thicknesses for acceptance purposes as well.

Some agencies specify spread rates (yield) in lieu of thickness. Spread rates can be checked as a quality-control measure by calculating the weight of material placed over a measured area. If measurement of spread rates is required as part of the quality-control plan, a technician can keep a log of beginning and ending stations for each truck delivered, along with the weight of each load. On this log, running values of spread rates can be calculated and reported to the paving superintendent on a periodic basis throughout the day.

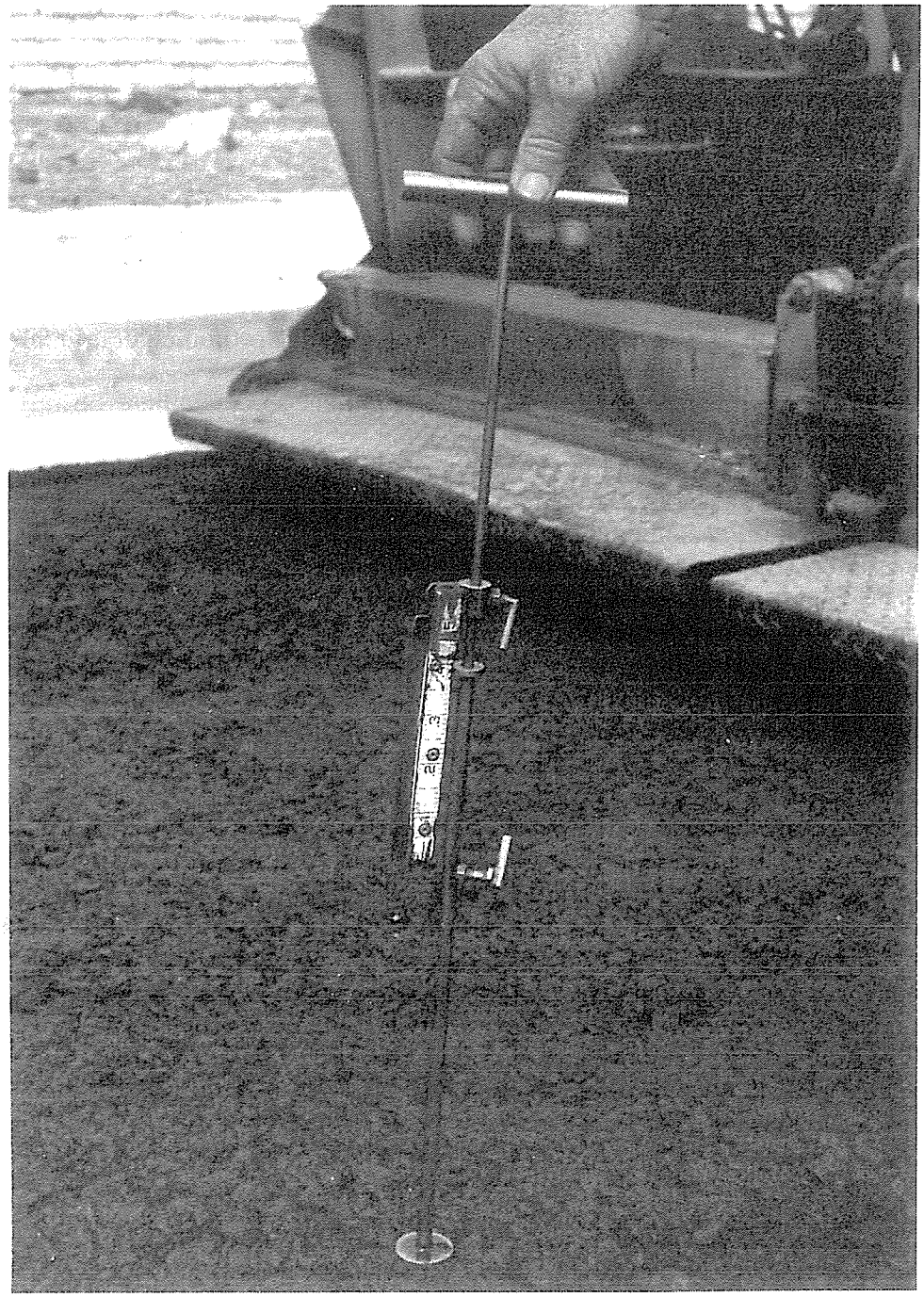


Figure 10.13 Use of Depth Gauge to Measure Loose Mat Thickness